

Session 8: Exercises (participant version)

Foundations of Clinical Research · Session 8 (Week 8, Saturday): Publish (finish the paper, get it out, present)

Part	Exercise	When	Time
A	Writing exercises: fix the spin, verb ladder, results sentence, title from PICO	warm-up / homework	15 min
B	[L2] Manuscript workshop 2: Introduction, Discussion, Abstract, Title	0:30	30 min
C	Getting published: spot the predatory journal; responding to reviewers	at home	25 min
D	Student appraisal pack: "Be the reviewer"	1:40	20 min

Part A: Writing exercises (section 8.1; warm-up or homework)

Use these to practise before or after Manuscript workshop 2: A2 and A3 can be done in class if a group finishes early; the rest are homework.

A1. Fix the spin

Rewrite each sentence so it says only what the data and the design allow.

#	Original sentence (from the papers)	Your rewrite
a	Paper 1 (p.10), primary outcome $p = 0.42$, difference 13.2 ml/kg (95% CI -15.2 to 41.6): the authors base the case for a larger trial on "the non-significant trend in fluid balance reduction".	
b	Paper 2 (p.4): "a positive fluid balance on the first ICU day did not impact the 28-day mortality".	
c	Paper 2 (title): "Impact of a cumulative positive fluid balance during the first three ICU days in patients with sepsis ..."	
d	Paper 1 (p.10): the protocol obtained "the lowest fluid balance on Day 5 ever published".	

A2. Climb the verb ladder

Choose the strongest verb each design allows: **was observed in** · **is associated with** · **may contribute to** · **reduced / did not reduce** · **causes**.

#	Study	Strongest honest verb
a	A case series of 12 ICU patients who developed pressure injuries under a new mattress	
b	A cohort of 3,000 septic patients: positive fluid balance and mortality, adjusted for severity	
c	A single randomised trial of 48 patients: restrictive protocol and day-5 fluid balance	
d	Five large randomised trials and a meta-analysis, all consistent	
e	A cross-sectional survey of hand-hygiene compliance on two wards	

A3. Write the results sentence (number + CI, no adjectives)

Write one sentence for each, in the style "In [whom], [outcome] was [effect] ([measure] [value], 95% CI [lower] to [upper])".

1. Paper 2, Day-3 cohort: 28-day mortality, positive vs negative fluid balance, HR 1.56, 95% CI 1.10 to 2.22, P = 0.012.
2. Paper 1: death by day 28, 6/24 (restrictive) vs 6/24 (control), relative risk 1.0, 95% CI 0.38 to 2.66.
3. Paper 2, Day-1 cohort: HR 1.17, 95% CI 0.85 to 1.60, P = 0.354.

A4. A title from your PICO

Formula: **[Design] of [Intervention / exposure] vs [Comparator] in [Population]: effect on / association with [Outcome]**. 10–16 words, design stated, no "impact", "novel", "first".

Our PICO: P _____ I/E _____ C _____ O _____

Working title: _____

Word count: ____ Design named? ☐ Calibrated verb (association for observational designs)? ☐

Part B: [L2] Manuscript workshop 2: complete the draft (30 minutes)

Start from last week's file (FLUID-ICU_Q[n]_draft1). Today you add the **Introduction, Discussion, Abstract and Title**, and the draft is complete.

Roles and timing

Minute	Who	Task
0–8	2 people	Introduction: three paragraphs (known → not known → aim), ~350 words, references from Sessions 1–2 + Papers 1–2

Minute	Who	Task
0–15	3 people	Discussion: six moves (answer, compare, explain, limitations with direction, implications, conclusion), ~600 words
0–15	1 person	Number and verb checker: reads Methods + Results again; prepares the abstract numbers
15–23	all	Abstract: Background / Methods / Results / Conclusions, ≤ 250 words, numbers copied from the Results
23–27	all	Title (10–16 words, design stated, calibrated verb) + 3–6 keywords (MeSH, not already in the title)
27–30	checker	Verb check: change every "cause", "impact", "effect of" and "show" to "associated with". Make every number identical in abstract and Results.

Discussion: six moves checklist

- [] **1 Answer:** first sentence answers the question with the main number (adjusted estimate + 95% CI)
- [] **2 Compare:** with Paper 2 and at least one other study, and why results may differ
- [] **3 Explain:** one short paragraph on mechanism / plausibility
- [] **4 Limitations:** observational design (residual confounding, especially severity), simulated/teaching data, missing data, single time point... each with its likely *direction*
- [] **5 Implications:** for practice (cautious) and research (e.g. a randomised trial such as Paper 1)
- [] **6 Conclusion:** one sentence, same verb and message as the abstract and title

Group-specific reminders

- **Q1:** crude and adjusted ORs differ. Explain confounding by indication in move 3/4.
- **Q2:** AKI may also *cause* fluid accumulation (reverse causation). Say so.
- **Q3:** ventilator-free days are skewed and include death as zero. Explain what the measure means.
- **Q4:** compare complete-case and multiple-imputation estimates; say what the missing-data assumption (MAR) means for your conclusion.
- **Q5:** survivors only. Those who died are excluded, so say who the result applies to.

Deliverable

A complete draft (~2,000 words, Template: `Templates/Manuscript_Template_IMRaD.docx`) saved as `FLUID-ICU_Q[n]_draft2`. After the course: peer-review swap (2 weeks) using the 10-question checklist (Part D) and a response letter (`Templates/Response_to_Reviewers_Template.docx`).

Part C: Getting published (section 8.2, at home)

C1. Spot the predatory journal

1. Circle the red flags in this (invented) invitation. How many can you find?

> **From:** editorial.board.gjcmr@freemail.com

> **Subject:** Invitation to publish in our esteemed journal!!

>

> Dear Respected Professor/Doctor,

>

> Greetings! We have gone through your excellent article and are delighted to invite you to submit to the *Global Journal of Clinical Medicine, Nursing, Engineering and Management Research*. We publish all areas of science.

>

> Our journal has an Impact Factor of 9.2 (International Universal Index). We guarantee peer review within 48 hours and publication in 5 days. Processing charges will be communicated after acceptance. Special discount this week only!

>

> Kindly submit your manuscript by replying to this e-mail.

2. Which journal would you trust? For each (invented) profile, write *Trust* / *Check further* / *Avoid* and one reason.

Journal profile	Your verdict	Reason
J1: Indexed in MEDLINE (confirmed in the NLM Catalog), editor names verifiable at universities, fee (APC) shown on the website, median 9 weeks to first decision, uses CONSORT/STROBE checklists.		
J2: Says "indexed in Google Scholar and Index Copernicus", promises acceptance in 72 hours, no fee information, editorial board photos without affiliations.		
J3: Open access, listed in DOAJ, small society journal, no impact factor yet, clear peer-review and retraction policy.		
J4: Its name differs by one word from a well-known journal; the website address is different from the real journal's; it e-mailed you first.		

3. Three checks you can always do: Think.Check.Submit checklist · DOAJ (open-access journals) · NLM Catalog ("currently indexed in MEDLINE"). You can also ask your librarian.

C2. Responding to reviewers

Imagine your group's study has been completed and submitted. Write a response row for each comment (thank, answer, say exactly what changed). One of them deserves a polite rebuttal.

#	Reviewer comment	Our response	Change made (page, line)
R1.1	"The title says your intervention <i>improves</i> outcomes, but this is an observational study."		
R1.2	"Please report the main result with a confidence interval, not only the p-value."		
R1.3	"Text and Table 2 give different numbers of patients analysed (212 vs 208)."		
R2.1	"Please add 1-year mortality." (Your protocol followed patients to day 28 only.)		

Tone checklist: ☐ thanked the reviewer ☐ answered every point ☐ said where the manuscript changed ☐ disagreed only with evidence, politely

Part D: Student appraisal pack ("Be the reviewer") (20 minutes, 1:40)

What you are doing

Two published papers answer our running question (**fluids in septic shock**) with different designs:

	Paper	Design	Who appraises it
Paper 1	Boulet N, et al. <i>Crit Care</i> 2024;28:429 (restrictive strategy targeting non-resuscitative fluids, OPTIFLUID)	Randomised controlled pilot trial, 48 patients	Groups 1, 2, 3
Paper 2	Hyun D-G, et al. <i>Ann Intensive Care</i> 2023;13:105 (cumulative positive fluid balance in the first three ICU days)	Propensity-score-matched cohort from a sepsis registry, 2516 patients	Groups 4, 5

You do **not** need to understand every statistic. Answer each question **Yes / No / Can't tell** and write **the page** where you found the answer. "Can't tell" is a good answer when the paper does not say.

> **Page numbers** are the journal's own page numbers printed at the top of each page ("Page X of 13" for Paper 1, "Page X of 9" for Paper 2). Use the original (unannotated) PDF or printout.

Step by step (10 minutes of group work)

Minute	What to do
0–1	Choose a timekeeper and a speaker . Split into three pairs: pair A = questions 1–3, pair B = 4–6, pair C = 7–10.
1–7	Each pair reads only the pages listed for its questions (table below) and fills in its rows of the grid.
7–9	The group goes through all 10 questions together and agrees the answers.
9–10	Write your one-sentence verdict ("This paper shows / suggests ... but ...").

Then the plenary (8 minutes): each group reports for 1 minute on the questions the instructor assigns, and we compare what each design may claim.

> **Tip:** read the papers' abstracts at home during the week before (Session 7 homework). The 10 minutes then go much further.

> **After the course** you use the **same 10 questions** to review another group's FLUID-ICU manuscript (peer-review swap, 2 weeks).

Where to look

Question	Paper 1 (RCT): pages	Paper 2 (cohort): pages
1 Question	Abstract (p.1); last paragraph of Background (p.3)	Abstract (p.1); end of Background (p.2)
2 Design	Title (p.1); "Trial design and oversight" (p.3)	Title (p.1); Methods (pp.2–3)
3 Participants	"Patients" (p.3); flow chart Fig. 1 (p.5); Discussion limitations (p.10)	"Data source and patient selection" (p.2); Results first paragraph (p.3)
4 Comparison groups	"Randomization and blinding" (p.3); Table 1 (p.6)	"Propensity score matching" (pp.2–3); Table 1 (pp.5–6); text p.4
5 Bias	"Randomization and blinding" (p.3); Discussion (p.8)	Methods (pp.2–3); Discussion limitations (pp.7–8)
6 Outcome	"Outcomes" (p.4)	"Propensity score matching and outcomes" (p.3)
7 Numbers	"Primary outcome" (p.6); Table 3 (p.10)	Abstract (p.1); "Outcomes" (p.4)
8 Words vs design	Abstract conclusion (p.1); Discussion (pp.8, 10); Conclusions (p.11)	Title (p.1); Abstract conclusion (p.1); Discussion (pp.4, 7); Conclusions (p.8)
9 Limitations	Discussion (pp.8–10)	Discussion (pp.7–8)
10 Use	Your judgement (whole paper)	Your judgement (whole paper)

Answer grid

Group number: _____ Paper: 1 / 2 Speaker: _____

#	Question (simplified appraisal checklist)	Yes	No	Can't tell	Page	Notes (one line)
1	Question: Can you write the study's PICO in one line?	☐	☐	☐		P: ... I/E: ... C: ... O: ...
2	Design: What is the design, and is it the right one for the question?	☐	☐	☐		
3	Participants: Who was included, and who was left out? Would your patients have been included?	☐	☐	☐		
4	Comparison groups: Were the groups similar at the start (Table 1)? How was this achieved (randomisation / matching / adjustment)?	☐	☐	☐		
5	Bias: Who knew which group patients were in (blinding)? Could that change the outcome?	☐	☐	☐		
6	Outcome: Was there ONE pre-specified primary outcome, measured the same way in both groups?	☐	☐	☐		
7	Numbers: What is the main result as an effect size with 95% CI? Is the CI narrow or wide?	☐	☐	☐		Effect: 95% CI:

#	Question (simplified appraisal checklist)	Yes	No	Can't tell	Page	Notes (one line)
8	Words vs design: Do the conclusions match the design ("associated" vs "caused"; no "trend" for $p = 0.42$)?	☐	☐	☐		Words found:
9	Limitations: Are the main limitations stated honestly? Name one the authors missed.	☐	☐	☐		Missed:
10	Use: Would this change practice in your unit? Why / why not?	☐	☐	☐		

Our verdict in one sentence: _____

One thing we would copy into our own proposal: _____

Hints if you are stuck

- **Q1:** the PICO is usually in the last paragraph of the Introduction ("we aimed to ..."). Is the paper's outcome the same as in our running question (28-day mortality)?
- **Q4:** a randomised trial makes groups similar by chance; a cohort can only match or adjust for what was measured. Find the sentence that says which.
- **Q5:** who decided how much fluid each patient received, and did they know the group?
- **Q7:** look for "difference" or "HR" followed by brackets. Does the interval include 0 (for a difference) or 1 (for a ratio)?
- **Q8:** search for the words *impact*, *show*, *trend*, *ever*, *regardless*.